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***Abstract*—Biometric authentication has offered a convenient and strong user bound mechanism for digital identity. But conventionally, the system remains fundamentally dependent of reusable biometric template compromise of which can be irreversible and cause user identity leakage forever. This paper proposed an end-to-end post-quantum biometric authentication framework that avoids transmitting reusable biometric templates while integrating user verification, post-quantum authentication and quantum-secured session establishment within a single protocol. The proposed architecture combines local biometric reconstruction, ML-DSA-based terminal and user authentication, signed QAuth session authorization, T12 biased-basis Quantum Key Distribution, recursive Cascade reconciliation, privacy amplification, HKDF-based session-key derivation and AES-GCM protected communication. A complete executable prototype, including a Gradio-based graphical interface, is developed together with a distance-aware physical-layer simulation incorporating fibre attenuation, detector inefficiency, intrinsic optical error and dark counts. The framework is evaluated through a 6,800-trial One-Factor-at-a-Time study across communication distance, basis bias, parameter estimation, QBER threshold, transmission batch size, adversarial interception and privacy-security margin.**

***Index Terms*—Post quantum cryptography, Quantum Key Distribution, Biometric authentication, Biometric Reconstruction**

I. Introduction

Biometric authentication has become a cornerstone of modern digital identity, replacing traditional credentials such as passwords and smart cards in applications ranging from banking and healthcare to border control and national digital infrastructure. Physiological traits such as fingerprints, facial features, and iris patterns provide a strong association between identity and authentication because they are inherently tied to an individual and cannot be forgotten or easily shared. However, this permanence also creates a significant security challenge. Conventional biometric systems store digital biometric templates which, although typically encrypted during storage and transmission, must eventually be reconstructed for matching. As a result, compromise of the template database can permanently expose a user's biometric identity, since unlike passwords, biometric traits cannot simply be revoked or replaced.

Recent advances in biometric cryptography, particularly fuzzy extractors, offer a promising alternative by reconstructing stable cryptographic secrets directly from noisy biometric observations while tolerating natural acquisition variations. This approach substantially reduces reliance on reusable biometric templates, yet securing the biometric alone is insufficient. The communication infrastructure remains vulnerable to both classical and emerging quantum attacks. Classical public-key mechanisms such as RSA and Elliptic Curve Cryptography (ECC) are expected to become insecure against large-scale quantum computers through Shor's algorithm, motivating the adoption of Post-Quantum Cryptography (PQC). Standardized lattice-based digital signature schemes such as ML-DSA provide quantum-resistant authentication, whereas Quantum Key Distribution (QKD) enables information-theoretically secure key establishment by exploiting quantum mechanical principles and detecting eavesdropping. However, QKD requires an authenticated classical channel to prevent man-in-the-middle attacks, while PQC alone does not provide secure key distribution, making the two technologies complementary rather than independent solutions.

Although substantial progress has been made in biometric cryptography, post-quantum authentication, and quantum-secured communication, these technologies are typically studied in isolation or integrated only partially. Few practical architectures simultaneously provide template-less biometric verification, quantum-resistant device authentication, authenticated quantum key establishment, and secure application-layer communication while accounting for realistic implementation factors such as optical attenuation, detector inefficiency, reconciliation leakage, and protocol latency. Consequently, developing a unified end-to-end authentication framework that bridges these components remains an important research challenge.

To address this gap, this paper proposes an integrated post-quantum biometric authentication framework that combines ML-DSA-based mutual authentication, biometric key reconstruction using fuzzy extraction principles, authenticated QAuth session tokens, T12 biased-basis Quantum Key Distribution, recursive Cascade-based key reconciliation, privacy amplification, HKDF-based session key derivation, and AES-GCM protected communication within a single protocol pipeline. The complete architecture is implemented as a functional research prototype with a graphical interface for end-to-end execution and evaluated under realistic physical-layer conditions including optical attenuation, detector efficiency, dark counts, and distance-dependent communication latency.

II. Related Work

Existing approaches toward quantum-resilient biometric authentication can be broadly viewed through four overlapping directions: biometric template protection, post-quantum authentication, quantum-secured communication, and hybrid architectures combining these mechanisms. The following discussion progressively examines these directions and identifies the gap addressed by the proposed framework.

A. Biometric Systems

Biometric authentication binds identity to physiological characteristics, but the permanence of these characteristics creates a security problem fundamentally different from conventional credentials: a compromised biometric cannot simply be replaced. Consequently, biometric information protection requires properties including confidentiality and renewability/revocability during storage and transfer [1]. Template-protection techniques such as cancellable biometrics, fuzzy commitments and biometric cryptosystems attempt to reduce this exposure. In particular, fuzzy extractors demonstrated that reproducible cryptographic material can be derived from noisy biometric observations using non-secret helper information [2]. More recent decentralized approaches combine fuzzy commitments with blockchain to authenticate users without directly disclosing biometric features [3]. These techniques reduce template exposure, but do not independently address the quantum vulnerability of the cryptographic communication and authentication mechanisms surrounding the biometric subsystem.

B. PQC-Based Authentication

The threat posed by sufficiently capable quantum computers has motivated migration from RSA and ECC toward post-quantum primitives. NIST standardized ML-DSA in FIPS 204 as a module-lattice-based digital signature scheme intended to remain secure against adversaries possessing large-scale quantum computers [4]. Within biometrics, Bauspieß *et al.* demonstrated post-quantum secure two-party computation for iris-template protection, showing that biometric processing itself can be redesigned around quantum-resistant cryptographic assumptions [5]. Nevertheless, PQC remains computationally secure: it protects authentication and communication by replacing vulnerable mathematical assumptions rather than providing information-theoretic key establishment. This motivates architectures in which PQC authenticates communicating entities while quantum communication provides an independent mechanism for session-secret establishment.

C. Quantum Communication

Quantum Key Distribution addresses this complementary requirement by allowing communicating parties to establish secret key material while making interception statistically observable through disturbance of transmitted quantum states. BB84 established the fundamental prepare-and-measure model for QKD [6], while Lo, Chau and Ardehali later showed that asymmetric basis-selection probabilities can substantially improve sifting efficiency without requiring uniformly selected bases [7]. Practical QKD additionally requires authenticated classical post-processing, information reconciliation and privacy amplification; Cascade provides an interactive mechanism for reconciling residual disagreements [8], while privacy amplification reduces an adversary's potential information about the resulting key [9]. QKD therefore provides a strong foundation for secret establishment, but does not by itself solve biometric identity verification or the authentication of the classical communication channel.

D. Current Hybrid Systems

Recent work has consequently begun combining PQC, quantum communication and biometric authentication. Qasim *et al.* proposed a decentralized biometric architecture in which ML-KEM first establishes a post-quantum shared secret, followed by QSDC-inspired authentication and decoy qubits for quantum-layer mutual authentication. A subsequent QKD stage derives symmetric key material that is used with AES to encrypt and transmit the extracted biometric template to a remote verification server [10]. Their work demonstrates the feasibility of layering post-quantum and quantum mechanisms around biometric communication, but the biometric template remains an object that must traverse the network and subsequently be processed against a biometric database.

The present work follows a different design principle: *the biometric should establish cryptographic identity rather than become the transmitted secret*. Biometric processing and ML-DSA credential reconstruction are therefore confined to the trusted terminal, while the server receives only cryptographic evidence of successful authentication. Device authentication, biometric-derived user authentication and QKD authorization are further separated through mutually authenticated ML-DSA exchanges and the signed QAuth mechanism. Only after these trust stages succeed is biased-basis QKD initiated, followed by reconciliation, privacy amplification and independent session-key derivation. The resulting architecture shifts the role of quantum communication from protecting a biometric template in transit to protecting the authenticated communication session established after biometric verification.

III. Proposed System Architecture

The proposed framework is designed as an end-to-end authentication and secure communication architecture that integrates biometric authentication, post-quantum cryptography and quantum key distribution into a single coherent protocol. Rather than treating these technologies as independent security mechanisms, the proposed system assigns each component a well-defined responsibility within the authentication pipeline. Biometric reconstruction establishes the identity of the user, ML-DSA authenticates every participating entity against quantum-capable adversaries, while Quantum Key Distribution securely establishes a fresh session key for subsequent encrypted communication. The resulting architecture eliminates the dependence on reusable biometric templates,

prevents unauthenticated participation in the quantum communication process and establishes a continuously protected classical communication channel.

Designing such an integrated protocol requires clearly defining the responsibilities and trust assumptions associated with every participating entity. Before discussing the protocol itself, it is therefore necessary to specify the capabilities, privileges and security assumptions associated with each participant. These assumptions collectively define the threat model under which the proposed system operates.

A. Threat Model

The proposed authentication framework consists of four principal entities: the user, the terminal, the authentication server and a potential adversary as shown in Table I. Each entity possesses different computational capabilities, privileges and security responsibilities. The threat model presented in this work explicitly defines the level of trust associated with every participant together with the assumptions under which the proposed protocol guarantees security.

The adversary in this work follows the Dolev - Yao communication model over the classical channel, possessing complete control over message transmission including interception, modification, replay and fabrication of network packets. In addition, the adversary is assumed to possess quantum computational capabilities and may perform probabilistic intercepts like resend attacks over the quantum communication channel. However, the adversary cannot compromise the trusted execution environment of the authentication terminal, reconstruct biometric information solely from publicly available helper data, or violate the physical principles governing quantum measurement and the no-cloning theorem.

Under these assumptions, the objective of the proposed framework is to simultaneously guarantee authenticated device communication, secure biometric verification, quantum-secure session establishment and confidential application-layer communication while preventing replay attacks, man-in-the-middle attacks, biometric template leakage and passive eavesdropping.

B. Overall Architecture

Figure 1 illustrates the overall workflow of the proposed authentication framework. The architecture is designed around two distinct operational phases, namely *user enrollment* and *user authentication*. While enrollment is performed only once for each user, the authentication phase is executed whenever the user requests access to a secure communication session. Separating these two phases minimizes unnecessary biometric processing while ensuring that long-term user credentials remain securely associated with a single enrollment procedure.

Every authentication session begins with mutual authentication between the terminal and the authentication server using ML-DSA digital signatures. This initial exchange establishes trust between the two communicating entities before any user-specific information is processed. By authenticating the participating devices prior to biometric verification, the framework prevents malicious terminals or impersonating servers from participating in the protocol, thereby eliminating a major source of man-in-the-middle attacks before the user authentication process begins.

If the user is not already registered within the system, the protocol enters the enrollment phase. The terminal collects the user's identity information together with an iris scan, from which a stable cryptographic secret is reconstructed using fuzzy extraction principles. This reconstructed secret is subsequently used to derive the user's ML-DSA key pair without permanently storing the biometric template itself. Upon successful enrollment, only the user's public verification key together with helper data required for future biometric reconstruction is transmitted to the authentication server. Since the helper data alone is insufficient to reconstruct the original biometric information, leakage of the server database does not directly expose the user's biometric identity.

For previously enrolled users, the protocol follows the authentication branch shown in the figure. After receiving the user's identity, the terminal retrieves the corresponding public verification key and helper data from the authentication server. A fresh iris observation is then used together with the helper data to reconstruct the user's private signing capability locally at the terminal. The reconstructed credential signs a challenge generated by the terminal, thereby proving possession of the user's biometric identity without transmitting the biometric template or reconstructed private key outside the trusted execution environment.

Successful biometric verification enables the terminal to generate a signed Quantum Authentication (QAuth) token containing the authenticated user's identity together with session metadata including timestamps and validity information. The server verifies the integrity and authenticity of the received token before initiating quantum communication. Consequently, Quantum Key Distribution is executed only after both the participating devices and the user have been successfully authenticated.

Following successful QAuth verification, the terminal and authentication server establish a shared secret using the T12 biased-basis Quantum Key Distribution protocol. The generated raw quantum key undergoes authenticated Cascade-based information reconciliation followed by privacy amplification to remove any information potentially available to an adversary. Finally, the reconciled secret is expanded using the HKDF key derivation function to generate independent symmetric encryption keys, which are subsequently employed by the AES-GCM protocol to establish a confidential and authenticated application-layer communication channel between the terminal and the server.

Although Figure 1 presents the proposed framework as a sequence of high-level protocol stages, each stage represents an independent cryptographic subsystem with distinct security objectives, computational requirements and communication responsibilities. Section 4 therefore examines each subsystem individually, discussing its mathematical formulation, implementation methodology and contribution to the overall security of the proposed architecture.

TABLE I
THREAT MODEL OF THE PROPOSED AUTHENTICATION FRAMEWORK

Entity	Role	Capabilities	Trust Assumption
User	Initiates authentication	Provides biometric information and requests secure communication	Considered honest but unable to securely store long-term cryptographic secrets.
Terminal	Trusted authentication device	Captures biometric data, reconstructs user credentials, performs ML-DSA operations and establishes QKD sessions	Trusted execution environment. Biometric information never leaves the terminal.
Authentication Server	Identity verification and communication endpoint	Stores public credentials, helper data and validates authentication requests	Trusted authority responsible for maintaining credential integrity and protocol state.
Adversary	External attacker	Can eavesdrop, intercept, replay, modify and inject classical communication. May partially intercept quantum transmissions and possesses quantum computational capabilities.	Cannot compromise the trusted execution environment of the terminal or violate the fundamental laws of quantum mechanics.

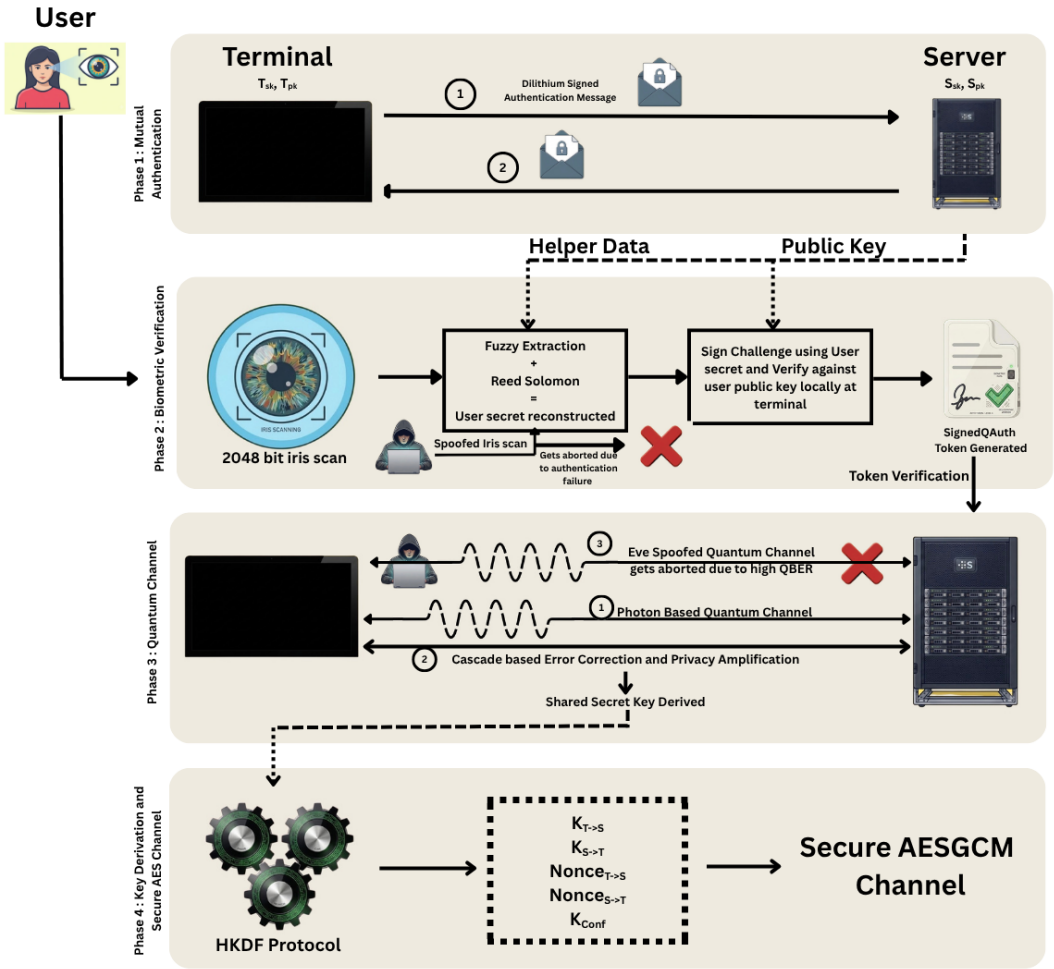


Fig. 1. Overall System Architecture for QBioAuth Pipeline

IV. PROTOCOL DESIGN AND IMPLEMENTATION

This section presents the implementation details of the proposed framework. Following the architectural workflow introduced in the previous section, each subsystem is described independently, highlighting its objective, protocol execution and interaction with subsequent stages. Together, these subsystems form the complete end-to-end authentication and secure communication pipeline.

A. System Initialization and Terminal-Server Mutual Authentication

Every authentication session begins by establishing trust between the terminal and the authentication server before any user-specific information is communicated. During system initialization, both entities are provisioned with independent ML-DSA public-private key pairs that uniquely identify them throughout the protocol. The corresponding public keys are exchanged during deployment and remain available for subsequent signature verification.

At the beginning of every session, the terminal generates an authentication request containing a timestamp and a nonce to prevent replay attacks. The request is digitally signed using the terminal's private ML-DSA key using Equation 1 and transmitted to the authentication server. Upon reception, the server verifies the signature using the terminal's registered public key using Equation 2. A successful verification authenticates the terminal and confirms both the integrity and freshness of the received message.

$$\sigma_T = \mathrm{Sign}_{sk_T}(m \,\|\, t \,\|\, n) \tag{1}$$

$$\mathrm{Verify}_{pk_T}(m, \sigma_T) = 1 \tag{2}$$

where:

- m is the authentication message,
- t is the timestamp,
- n is the nonce,
- σ_T is the terminal signature,
- sk_T is the terminal private key, and
- pk_T is the terminal public key.

Following successful terminal authentication, the server performs the same procedure in the reverse direction. The server generates its own signed authentication challenge, which is subsequently verified by the terminal using the server's public verification key. Only after both signatures have been successfully verified does the protocol consider the communication endpoints mutually authenticated and allow the authentication process to proceed.

This initial mutual authentication stage establishes a trusted communication relationship between the participating devices before any biometric information is processed. Consequently, subsequent enrollment, user authentication and quantum key establishment are performed only after both communication endpoints have been successfully verified.

B. User Enrollment

User enrollment is performed only once for each participant and establishes the long-term cryptographic identity associated with the user's biometric information. During enrollment, the terminal collects the user's identity information together with an iris scan. Unlike conventional biometric systems, the proposed framework does not store or transmit the biometric template itself. Instead, the terminal derives a stable binary iris representation that serves as the input to the fuzzy reconstruction process.

The reconstructed biometric secret is subsequently used to deterministically generate the user's ML-DSA key pair. The corresponding public verification key, together with the helper data required for future biometric reconstruction, is securely transmitted to the authentication server and associated with the user's identity. Since the helper data alone does not reveal sufficient information to reconstruct the original biometric, compromise of the server database does not directly expose the user's iris information.

Upon successful enrollment, the server permanently stores only the user's public verification key and helper data, while the terminal discards all temporary enrollment variables. Consequently, subsequent authentication sessions reconstruct the user's signing capability directly from a fresh biometric observation rather than relying on a permanently stored private key.

C. User Authentication

Whenever an enrolled user requests access to the system, the authentication process begins by retrieving the corresponding public verification key and helper data from the authentication server using the user's registered identity. A fresh iris scan is then acquired by the terminal and converted into its binary representation. Owing to natural acquisition variations, the observed iris code is expected to differ slightly from the enrolled representation. The fuzzy reconstruction procedure therefore combines the new observation with the stored helper data to recover the user's cryptographic secret while tolerating limited biometric noise.

Following successful reconstruction, the recovered secret regenerates the user's ML-DSA private signing key locally within the trusted terminal. The terminal subsequently generates a fresh authentication challenge and requires the reconstructed credential to produce a valid digital signature. The server verifies the received signature using the user's stored public verification key. Successful verification simultaneously confirms the authenticity of the user and the correctness of the reconstructed biometric secret without requiring the biometric template or private key to be transmitted over the communication channel.

Only after successful user authentication does the protocol proceed to session establishment through the generation of an authenticated QAuth token, which serves as the authorization mechanism for initiating quantum key distribution.

D. QAuth Token Generation

Successful biometric authentication confirms the identity of the user but does not, by itself, authorize the initiation of a quantum communication session. To explicitly separate user authentication from session establishment, the proposed framework introduces a signed authorization mechanism referred to as the *Quantum Authentication (QAuth) Token*. The QAuth token serves as a cryptographically verifiable proof that the terminal has successfully authenticated the user and is requesting permission to establish a secure communication session.

Following successful verification of the user's reconstructed ML-DSA signature, the terminal generates a QAuth token containing the authenticated user's identity, a unique session identifier, a timestamp and a predefined validity period. The complete token is digitally signed using the terminal's ML-DSA private key before being transmitted to the authentication server. Since the terminal itself has already been mutually authenticated at the beginning of the protocol, the signed token acts as a trusted authorization request originating from a verified device.

Upon receiving the QAuth token, the authentication server verifies both the integrity of the token and the authenticity of the terminal's digital signature. The timestamp and validity interval are additionally checked to prevent replay of previously authorized sessions. Only after all verification steps have been successfully completed does the server authorize the execution of the Quantum Key Distribution protocol.

The generated QAuth token is represented as

$$Q = \{UID, SID, T_issue, T_{exp}, Status\}, \tag{3}$$

where UID denotes the authenticated user identity, SID represents the session identifier, T_issue is the generation timestamp, T_{exp} specifies the token expiration time and *Status* indicates successful terminal-side authentication. The terminal subsequently generates

$$\sigma_Q = \mathrm{Sign}_{sk_T}(Q), \tag{4}$$

which is verified at the server using the terminal's registered public verification key before session establishment.

The QAuth token therefore acts as the authorization bridge between biometric verification and quantum communication. By ensuring that only authenticated terminals acting on behalf of successfully verified users are permitted to initiate Quantum Key Distribution, the proposed framework avoids unnecessary quantum resource allocation while providing explicit session-level authorization.

E. Quantum Key Distribution for Session Establishment

Following successful verification of the QAuth token, the authentication server authorizes the establishment of a quantum communication session with the terminal. The proposed framework employs the T12 biased-basis Quantum Key Distribution protocol to generate a shared symmetric secret between the communicating entities. Unlike the conventional BB84 protocol, T12 intentionally biases basis selection towards the computational basis, thereby increasing the probability of basis agreement and improving secret key generation efficiency without altering the underlying security guarantees. The basis selection probabilities satisfy

$$P(Z) = p, \qquad P(X) = 1 - p, \tag{5}$$

where $p = 0.8$ in the validated baseline configuration of the proposed framework.

After transmission of the quantum states, both parties perform authenticated basis reconciliation over the previously established ML-DSA protected classical channel. Measurement outcomes corresponding to incompatible preparation and measurement bases are discarded, while the remaining sifted bits are retained for subsequent reconciliation. Parameter estimation is then performed over a predefined fraction of the sifted key to estimate the Quantum Bit Error Rate (QBER) according to Equation 6. If the estimated QBER exceeds the predefined security threshold, the current session is immediately aborted and the protocol initiates a fresh quantum transmission.

$$QBER = \frac{N_{\mathrm{error}}}{N_{\mathrm{compared}}}, \tag{6}$$

where N_{error} denotes the number of mismatched bits and N_{compared} represents the number of disclosed bits used for parameter estimation.

The observed QBER is influenced by multiple physical phenomena including optical attenuation (Equation 7), detector inefficiency, intrinsic optical misalignment and detector dark counts.

$$\eta_{\mathrm{fiber}}(d) = 10^{-\alpha d}, \tag{7}$$

These effects are jointly modelled throughout the implementation to emulate a realistic quantum communication environment rather than an idealized noiseless channel.

Quantum transmission continues until

$$N_{\mathrm{det}} \geq N_{\mathrm{target}}, \tag{8}$$

where N_{target} denotes the minimum number of detector events required before parameter estimation and key reconciliation are initiated.

The implementation further maintains a fixed set of protocol parameters governing basis selection probability, parameter-estimation fraction, target detector events, reconciliation configuration and privacy security margin during normal system execution. These parameters constitute the validated baseline configuration of the proposed framework and remain unchanged throughout end-to-end authentication unless explicitly modified during experimental evaluation.

The resulting sifted key represents a shared but imperfect secret due to the presence of channel noise and detector imperfections. Consequently, an additional information reconciliation stage is required to eliminate residual discrepancies before the secret can be safely employed for cryptographic applications.

F. Cascade-Based Information Reconciliation

The sifted key generated by the Quantum Key Distribution stage is expected to contain a small number of bit disagreements arising from channel attenuation, detector imperfections and background noise. Directly using this key for cryptographic communication would therefore result in inconsistent session keys at the terminal and the authentication server. To

eliminate these residual discrepancies, the proposed framework employs the Cascade information reconciliation protocol over the authenticated classical communication channel.

The Cascade protocol recursively partitions the sifted key into blocks and compares their parity values between the communicating parties. Whenever a parity mismatch is detected, a binary search procedure is performed within the corresponding block to identify and correct the erroneous bit while minimizing information leakage over the classical channel. Multiple reconciliation passes with varying block sizes are subsequently executed to correct errors that may remain undetected during earlier iterations.

Unlike a fixed-pass Cascade implementation, the proposed framework employs a recursive reconciliation strategy in which the protocol automatically reinitiates reconciliation whenever consistency verification fails after the configured reconciliation passes. This significantly improves end-to-end session reliability while preserving compatibility with the standard Cascade procedure.

For a block B_i containing n_i bits, the transmitted parity is computed as

$$P(B_i) = \bigoplus_{j=1}^{n_i} b_j, \tag{9}$$

where $\oplus$ denotes the bitwise XOR operation.

If the parity values computed by the terminal and the server differ,

$$P_T(B_i) \neq P_S(B_i), \tag{10}$$

the block is recursively divided until the erroneous bit is uniquely identified and corrected.

Information reconciliation terminates once

$$K_T = K_S, \tag{11}$$

where K_T and K_S denote the reconciled keys obtained by the terminal and the authentication server, respectively.

Although reconciliation ensures that both communicating parties possess identical secret keys, the exchange of parity information inevitably leaks a limited amount of information over the authenticated classical channel. Consequently, a privacy amplification stage is required to compress the reconciled secret into a shorter key that is computationally independent of all information potentially available to an adversary.

G. Privacy Amplification

Although the Cascade reconciliation protocol ensures that both the terminal and the authentication server possess identical secret keys, the exchange of parity information during reconciliation inevitably leaks a limited amount of information over the authenticated classical channel. Furthermore, an adversary may have acquired partial information through quantum channel observations prior to reconciliation. Consequently, the reconciled key cannot be directly employed for cryptographic communication and must first undergo a privacy amplification stage.

The proposed framework performs privacy amplification by applying a cryptographic hash function to the reconciled secret, thereby compressing it into a shorter session key whose security is independent of any partial information potentially available to an adversary. Since cryptographic hash functions are computationally one-way and exhibit strong avalanche properties, even small uncertainties regarding the reconciled key translate into negligible knowledge about the final compressed secret.

Unlike reconciliation, which attempts to preserve as much key material as possible, privacy amplification intentionally sacrifices key length in exchange for stronger security guarantees.

Let the reconciled secret generated after Cascade be denoted by

$$K_{PA} = SHA256(K_{REC}), \tag{12}$$

where K_{REC} represents the reconciled quantum secret.

The final secret length is selected according to

$$L_{PA} = L_{REC} - \lambda, \tag{13}$$

where L_{REC} denotes the reconciled key length and λ represents the security margin that compensates for information leakage during reconciliation and parameter estimation.

The resulting compressed secret satisfies

$$K_{PA} = K_{PA}^{(T)} = K_{PA}^{(S)}, \tag{14}$$

where $K_{PA}^{(T)}$ and $K_{PA}^{(S)}$ denote the privacy-amplified secrets independently computed by the terminal and the authentication server, respectively.

The privacy-amplified secret constitutes the final output of the quantum key establishment protocol. However, directly employing this secret for application-layer encryption would tightly couple the quantum communication stage with the encryption algorithm. The proposed framework therefore performs an additional key derivation stage to generate independent cryptographic session keys suitable for authenticated symmetric encryption.

H. HKDF-Based Key Derivation and Classical Channel Communication

Following privacy amplification, both the terminal and the authentication server possess an identical high-entropy quantum secret. Although this secret is suitable for cryptographic use, directly employing it as an encryption key tightly couples the quantum communication protocol with the application layer and limits future extensibility. The proposed framework therefore employs the HMAC-based Key Derivation Function (HKDF) to derive independent cryptographic session keys from the shared quantum secret.

HKDF expands the privacy-amplified secret into multiple independent cryptographic materials, each assigned a dedicated purpose within the communication session. In the

proposed implementation, separate keys are derived for AES-GCM encryption, initialization vector generation and message authentication. Consequently, compromise of one derived key does not reveal information regarding the remaining session keys or the original quantum secret.

The key derivation process is represented as

$$\{K_{TS}, K_{ST}, N_{TS}, N_{ST}, K_{CONF}\} = HKDF(K_{PA}, Salt, Info), \tag{15}$$

where K_{PA} denotes the privacy-amplified quantum secret, while K_{TS}, K_{ST}, N_{TS}, N_{ST} and K_{CONF} represent the terminal to server key, server to terminal key, terminal to server nonce base, server to terminal nonce base and a confirmation key to ensure that both parties have agreed on the exact same keys.

The derived encryption key is subsequently employed by the AES-GCM authenticated encryption algorithm to establish a secure classical communication channel between the terminal and the authentication server. For every transmitted message, AES-GCM simultaneously guarantees confidentiality, integrity and authenticity while generating an authentication tag that enables the receiver to detect any unauthorized message modification during transmission.

Application-layer communication is therefore performed as

$$(C, \tau) = AES\text{-}GCM_{K_{enc}}(M), \tag{16}$$

where M denotes the plaintext message, C represents the ciphertext and τ denotes the authentication tag generated during encryption.

The combination of privacy amplification, HKDF and AES-GCM completes the secure communication pipeline by transforming the reconciled quantum secret into application-ready cryptographic material. The resulting session enables authenticated and confidential communication while remaining cryptographically independent of the underlying quantum key establishment process.

I. End-to-End Protocol Pipeline and Graphical User Interface

To demonstrate the practical integration of the proposed authentication framework, all protocol stages described in the preceding subsections are sequentially organized into a unified end-to-end execution pipeline. The implementation orchestrates terminal–server mutual authentication, biometric enrollment and verification, QAuth token generation, T12 Quantum Key Distribution, Cascade-based information reconciliation, privacy amplification, HKDF-based session key derivation and AES-GCM secured communication as a single automated workflow. To improve usability and facilitate experimental evaluation, the complete pipeline is additionally exposed through a Gradio-based graphical user interface, enabling interactive execution of the protocol while preserving the underlying cryptographic implementation. The GUI provides a user-friendly interaction layer for the proposed framework while leaving the underlying protocol logic and cryptographic execution unchanged.

TABLE II
BASELINE CONFIGURATION OF THE PROPOSED FRAMEWORK

Parameter	Assumption / Role	Baseline
Z-basis probability	Probability of selecting the dominant T12 basis	80%
Parameter-estimation fraction	Fraction of sifted bits disclosed for QBER estimation	15%
QBER abort threshold	Maximum estimated QBER permitted before session rejection	8%
Target detector events	Minimum accumulated detections before QKD processing	8192
Transmission batch size	Optical pulses represented in each acquisition batch	8192
Source emission efficiency	Probability of successful photon emission	100%
Coupling efficiency	Efficiency of coupling photons into the optical channel	80%
Detector efficiency	Mean photon detection efficiency	20%
Intrinsic optical error	Mean baseline optical error probability	1%
Dark-count probability	Probability of a spurious detector event per gate	10^{-6}
Dark-count error probability	Probability that a dark-count detection produces an incorrect bit	50%
Cascade passes	Configured recursive reconciliation passes	6
Privacy security margin	Additional key-length deduction during privacy amplification	128 bits
Maximum QKD attempts	Maximum fresh QKD attempts following recoverable failure	3
Synthetic iris noise	Bit-error probability applied to a repeated synthetic iris scan	1%
Synthetic iris rotation	Circular displacement used to model acquisition misalignment	-4 bits
QAuth validity	Maximum authorization-token lifetime	120 s

TABLE III
PARAMETERS EVALUATED IN THE OFAT STUDY

Factor	Evaluated Values
Communication distance	0, 20, 50, 80, 100, 150, 200 km
Transmission batch size	1024, 2048, 4096, 8192, 16384 pulses
Z-basis probability	50%, 70%, 80%, 85%, 93.75%
Parameter-estimation fraction	3%, 5%, 10%, 15%, 20%
QBER abort threshold	3%, 5%, 8%, 10%, 12%
Eve interception fraction	0%, 10%, 25%, 50%, 75%, 100%
Privacy security margin	64, 128, 192, 256 bits

V. EXPERIMENTAL METHODOLOGY

The proposed framework was evaluated step by step in a controlled simulation experiment environment to determine how the behaviour of the protocol and trustworthiness of the results changes under varying physical conditions and introduction of noises.

A. Experimental Setup

All the experiments are ran using the same end-to-end implementation which has been details in Section IV. Unless varied as an experiment factor, the protocol and physical channel parameters are as described in Table II. This creates a common operation point from which the individual effects of selected parameters can be measured.

B. One-Factor-at-a-Time Experimental Study

To identify the behaviour of the proposed framework with controlled varying properties, a One-Factor-at-a-Time (OFAT) study was done. Through all the experiments, a single physical or protocol parameter was varied keeping all the other at the baseline config as mentioned in Table II. Using this approach, we could identify the changes in QBER, key generation time, reconciliation behaviour, transmission attempt requirements and protocol latency. We evaluated 7 factors across 27 exp. cases as summarised in Table III

The factors so that they include the effects of both the protocol layer and the physical layer of the environment. Communication distance and transmission batch size show the resource requirements due to optical attenuation. The Z-bases probability tests the efficiency gained through biased-basis QKD compared to the standard BB84 protocol. The parameter-estimation fraction and QBER threshold determine the trade-off between the key material retained and confidence in the quantum channel's security. The Eve interception fraction values evaluates te protocol's behaviour under stronger intercept resend attack on the system. The parameter ranges from no interception to complete interception of the quantum transmission. Lastly, the privacy security margin measures the cost of key compression after the reconciliation has been done and a final identical key material is available between the Terminal and the Server. Together, these factors inform us about the principal conditions expected to vary the expected results and the impact they create on security, reliability and the efficiency of the proposed quantum secured authentication pipeline.

We used a fixed master seed for reproducibility of the results. Each OFAT case was tested against 100 trials with Eve trying to intercept (100% interception) and 100 trials with Eve absent (0% interception) from the scenario except the 6 cases of Eve Interception Fraction leading to a total of 6800 individual records.

C. Evaluation Metrics

The OFAT experimental study we executed were evaluated against metrics ranging from the quantum channel, reconciliation process and end-to-end protocol success. The primary measurements include session success rate, quantum channel QBER, total transmitted pulses required to complete the process, sifted and reconciled bitstring length, leakage during the key correction, final privacy amplified secret length and end-to-end execution latency. Together, these evaluation results tell us if the affected parameter affects the *security, reliability* or *resource efficiency* in the current design.

Protocol reliability is measured through the session success rate,

$$S_{\text{rate}} = \frac{N_{\text{successful}}}{N_{\text{total}}} \times 100\%, \tag{17}$$

where $N_{\text{successful}}$ represents trials that successfully complete the required authentication, QKD, reconciliation and key-establishment stages. Quantum channel behaviour is characterised using QBER estimation while the total number of transmitted pulses and acquisition rounds measure the optical resources required to accumulate detector events on the receiver's end.

The efficiency of post quantum channel processing is evaluated using the length of reconciled key material, number of errors corrected in the shared secret and leakage during key correction. The latter records the classical information shared through Cascade parity exchanges and therefore reveals the security cost of correcting residual key disagreements. After the privacy amplification, the final secret retention ratio is determined by

$$R_{\text{secret}} = \frac{L_{\text{PA}}}{L_{\text{sifted}}}, \tag{18}$$

where L_{PA} and L_{sifted} denote the final privacy-amplified and initial sifted key lengths, respectively.

The end-to-end latency estimates the total time required to complete an authenticated session and to generate usable cryptographic material to set up a clean, threat free communication channel. The evaluation additionally records optical acquisition latency allowing us to separate the degradation caused by quantum channel and the overhead classical computation overhead introduced by authentication, reconciliation and key derivations. These measurements collectively provide a baseline to perform a comparative analysis presented in the following section.

VI. Security Analysis

The proposed framework doesn't rely on a single security mechanism. Instead, the security is established progressively allowing various attacks to be mitigated at the different stages of the pipeline. This section details the protection ensured at each trust transition and identifies the attacks mitigated by this protocol under the previously defined threat model in Table I.

A. Device and User Authentication Security

Terminal and server mutual authentication is established using the NIST standard ML-DSA dilithium signatures as the first step of the entire protocol. An adversary trying to attack the classical channel at this stage may intercept or modify the authentication messages. But it cannot generate a valid signature for either endpoint without the possession of the corresponding hardware secured private signing key. The mutual verification establishes the identities of both the devices before any biometric information or quantum resources are accessed.

After the device and the server are mutually authenticated, user authentication is a separate trust requirement. Possessing the access to the terminal alone can not guarantee the possession and authenticity of the user's identity. The user's ML-DSA signing ability if reconstructed from a fresh iris scan rather than kept as a stored private key. User authentication succeeds only when the reconstructed credential produces a signature verifiable under the associated private key. By this segregation of trust, we answer the two key questions separately, *where* the request originates and *on whose behalf* was the request made and is that person really there?

B. Biometric and Helper-Data Security

All the biometric information that needs raw information is processed locally at the trusted terminal and is never transmitted to the server. During the time of enrollment, the server only gets the access to the associated public key data of the dilithium key pair and the non secret public helper data required for the fuzzy reconstruction of the dilithium secret

key. Therefore, a fresh biometric scan must closely match to the scan at the time of the enrollment to reconstruct the same cryptographic secret.

The helper data is treated as an object of public information and is not assumed to provide confidentiality by itself. The security rests on the assumption that during the construction of the helper data at the time of enrollment, it should not expose sufficient information to reconstruct the enrolled biometric or the derived secret key without a sufficiently similar biometric observation. Following the logic, even if the server side authentication database gets compromised, it still lacks the ability to directly reveal a reusable biometric template or the user's ML-DSA private credential.

C. QAuth and Replay Resistance

After a successful biometric authentication, the terminal needs to inform the server that the user is authorised to make the request and is present at the time of request to initiate a quantum channel. The QAuth token binds the authenticated user to a fresh session identity, issuance time, expiration time and authentication state. The token is signed by the terminal and shared to the server. Any modifications to either of these fields invalidates the ML-DSA signature. The expiration timestamp and session state verification prevent a previously valid authorisation request message from being used indefinitely.

QAuth token is a security boundary between an established identity and quantum session establishment. An adversary must be able to access both a valid user authentication result and a valid terminal signature in the permitted session duraction before the server initiates a QKD channel and dedicates quantum resources blocking any unauthorised participants.

D. Quantum and Session-Key Security

Once the terminal sends the QAuth token and it is verified by the server, the server initiates a QKD session which establishes a secret material between the authenticated terminal and the server. Interception attacks or resend activity on the quantum channel introduces observable disturbances that raise the measured QBER. Sessions exceeding the configured QBER threshol of 8% are rejected while the accepted ones undergo Cascade based reconciliation followed by privacy amplifications to protect against the information leaked during parameter estimation and reconciliation.

The privacy amplified secret is subsequently processed using HKDF to produce independent direction specidic encryption keys, nonce derivation materials and key confirmation material. The separation prevents direct reuse of the quantum secret across cryptographic purposes and isolates communication in opposite directions. AES-GCM then provides authenticated encryption for application-layer messages, allowing unauthorized ciphertext modification to be detected by the receiving endpoint. The resulting construction therefore combines post-quantum endpoint authentication with QKD-derived session material rather than treating either mechanism as a substitute for the other.

TABLE IV
THREAT COVERAGE OF THE PROPOSED FRAMEWORK

Threat	Coverage	Primary Protection
Endpoint impersonation	Yes	Mutual ML-DSA authentication
User impersonation	Yes	Biometric reconstruction and ML-DSA verification
Classical MITM	Yes	Authenticated ML-DSA protocol messages
Replay attack	Yes	QAuth session identifier, timestamps and expiration
Biometric template interception	Yes	Local biometric processing; template is not transmitted
Server helper-data leakage	Partial	Helper data does not directly contain the biometric or private signing key
Passive quantum eavesdropping	Yes	QKD parameter estimation and privacy amplification
Intercept–resend attack	Yes	QBER-based disturbance detection and session rejection
Classical message modification	Yes	AES-GCM authenticated encryption
Quantum attack on classical authentication	Yes	ML-DSA post-quantum signatures
Compromised trusted terminal	No	Outside the defined threat model
Denial-of-service attack	No	Availability protection is outside the present protocol scope

The protections summarized in Table IV are conditional on the trust assumptions established in Section III. In particular, the framework does not claim security following compromise of the trusted terminal, nor does it attempt to guarantee service availability against denial-of-service attacks. These boundaries, together with implementation and simulation assumptions that constrain the present prototype, are discussed explicitly in Section VII.

VII. SYSTEM ASSUMPTIONS AND LIMITATIONS

The security of the current framework system is conditional on several deployment assumptions. The authenticating terminal is considered to be a trusted execution evironment whose Hardware Security Module or the hardware backed secret key can not be extracted, replicated, maliciously modified. The physical security of the terminal lies outside the scope of the proposed model.

The system also assumes the availability of an iris aqusition device capable of producing the required 2048-bit iris scan representation with a maximum of 3% noise allowing the fuzzy reconstruction of the biometric secret.

Once the credential and key reconstruction process is completed, the iris scan is immediately deleted from the accessible memory by the protocol. It is important that the garbage data of the operating system is cleaned up as well and a sensitive-data zeroisation protocol erases that data from the memory.

The present work should consequently be interpreted as an experimental research prototype rather than a production-ready biometric or QKD deployment.

Biometric acquisition is represented using synthetic 2048-bit iris codes and controlled noises rather than measurement from a real physical iris scanning hardware. Sensor specific effects and noises are not directly evaluated as a part of this study. Similarly, the quantum communication channel is simulated using a distance aware physical channel model incorporating the different noises rather than an experimental quantum channel. The framework also doesn't provide protection against hardware side channel attacks, denial of server attacks, malicious insider nodes or quantum repeaters. These

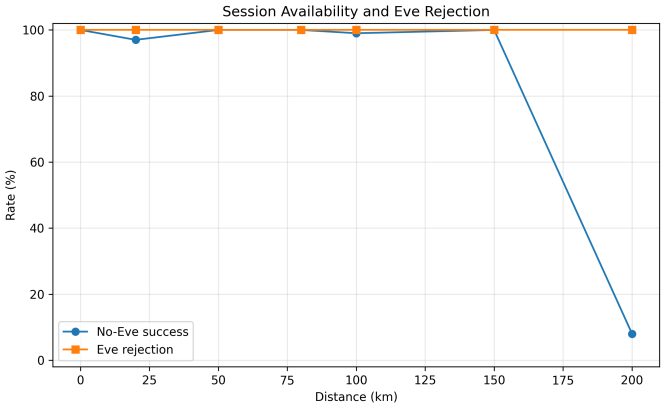


Fig. 2. Effect of Distance on Channel Success

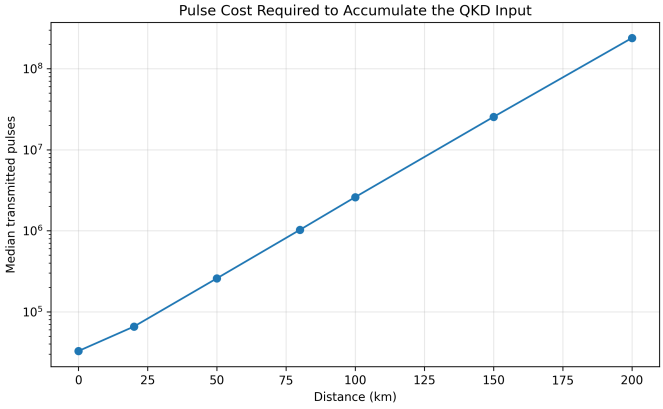


Fig. 3. Effect of Distance on the Number of Transmissions

limitations define the boundary within which the experimental results presented in this work should be interpreted.

VIII. Results and Discussion

The OFAT study conducted shows that the proposed framework runs stably around the baseline operating region. The behaviour of the entire protocol also changes predictably as the baseline metrics change when the physical channel loss, fraction of Eve intercepted channel and protocol specific parameters are increase.

A. Effect of Communication Distance

Out of all the factors that were studied as described in Table III, the increase in distance separation between the terminal and the server creates the strongest physical layer effect as shown in Fig. 2. In the absence of Eve, the protocol achieves a success rates of 100%, 97%, 100%, 100%, 99% and 100% at distances from 0 to 150 km, respectively. At 200 km, however, successful completion fell sharply to 8%. This is also accompanied by a sharp rise in the mean number of required transmitted pulses from 3.09×10^4 at 0 km to 2.54×10^7 at 150 km and 2.43×10^8 at 200 km as visualised in Fig. 3.

The results also show that attenuation is not directly linked to the QBER. The fibre attenuation according to Eq. 7 causes lesser number of photons to reach the detectors from the source. This forces the transmitter to increase the number of pulses to accumulate the minimum required detector events. However, at a sufficiently large distance, number of transmissions increase enough that the dark count event become non negligible causing the mean QBER to increase from 1% over 0–100 km to 1.29% at 150 km and 3.96% at 200 km which is visualised in Fig. 4

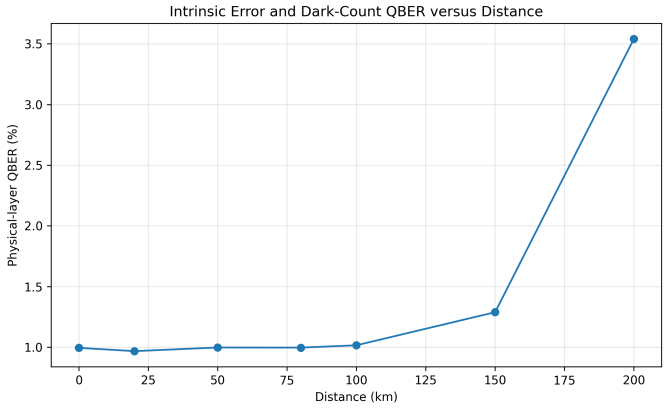


Fig. 4. Effect of Distance on Dark Counts

This behaviour directly affects uskable key generation. The mean privacy-amplified secret remained approximately 1.28–1.36 kbits up to 100 km, which decreases to 1.17 kbits at 150 km and collapses at 200 km, where only a small fraction of sessions completed successfully. The results therefore identify approximately 150 km as a practical upper operating region for the simulated baseline configuration rather than implying a strict theoretical distance limit.

B. Effect of Basis Bias in the QKD

The biased basis technique or the T12 revolves around the theory that a known increase in the probability of basis selection increases the probability of successful basis matching and key sifting after the qubits have been successfully delivered. This is confirmed by the experimental study as the basis-match rate increases from approximately 50.1% at $P(Z) = 0.5$ to 68.1% at the selected baseline of $P(Z) = 0.8$, reaching 88.2% at $P(Z) = 0.9375$. Consequently, the mean sifted-key length increased from approximately 2051 bits to 2788 bits at the baseline and 3614 bits under the strongest evaluated bias.

However, this technique's security vs efficiency trade off is also visible in the results. Under 100% intercepted channel, all sessions are rejected for $P(Z) \leq 85\%$. But almost 5% of the trails succeeded at $P(Z) = 93.75\%$. The mean estimated QBER at 100% Eve interception also decreased from 25.6% at unbiased basis selection to only 2.33% at the strongest bias. This shows that a biased basis though improves the key generation efficiency, it can still introduce a security lapse. The selected $P(Z) = 80\%$ baseline provides a more conservative compromise which retains the efficiency advantage of T12 while rejecting the possible security compromise.

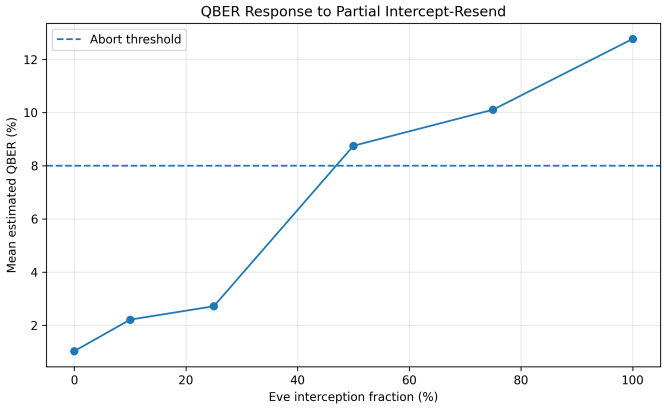


Fig. 5. Effect of Eve on QBER

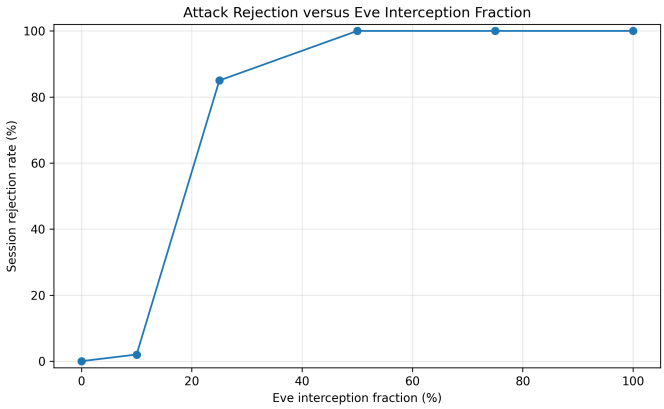


Fig. 6. Effect of Eve on Channel Success

C. Adversarial Interception Behaviour

THe Eve-fraction experiment shows the progressive degradation of the quantum channel as the probability of the intercepted photons in the channel increase. At 0% interception or no Eve present, all trials complete successfully with a mean estimated QBER of 1.03% due to the noises. At 10% interception, the mean QBER increases to 2.21% while the protocol still completes 98% of trials. At 25% of channel interception, only 15% of trials completed to success and no trial was able to set up a secure cryptographic session material at interception probability of 50%, 75% or 100% as we can see in Fig. 5 and Fig. 6.

An equally important effect occurs before complete session rejection. At 10% interception, successful trials retained an average privacy-amplified secret of approximately 729 bits, compared with 1312 bits without Eve. At 25% interception, this fell to approximately 49 bits. The reduction results from the larger number of disagreements that must be corrected and the corresponding increase in reconciliation leakage. Thus, adversarial disturbance is expressed not only through session rejection but also through a rapidly decreasing quantity of usable secret material in the subset of sessions that remain below the configured abort threshold.

D. Post-Processing and Protocol Parameters

Varying the size of transmission batch from 1024 to 16384 pulses doesn't introduce a major change in the number of pulses. Its primary effect is operational efficiency as the number of acquisition rounds decreases from 251 to 16. Batch size therefore controls the granularity rather than the optical efficiency.

Varying the parameter estimation fraction shows a clearer tradeoff between security and resources. Increase the disclosing fraction from 3% to 20% reduces the amount of key material left for further privacy amplification. Revealing higher fraction gives a larger statistical sample for QBER estimateion. Across the rest of the trials where the parameter estimation length is not varied, success remains approximately $99\% - 100\%$ which shows that the selected baseline of 15% operates comfortable within the stable region.

Similarly, varying the QBER abort threshold between 3% and 12% produced negligible differences under normal conditions because the underlying channel QBER remains near 1%. Complete intercept–resend attacks are rejected across all evaluated thresholds. The threshold therefore has a limited influence on the channel's performace while legitimate and adversarial QBER distributions remain well separated, although its importance increases under noisier channels or weaker interception.

Finally, increasing the privacy-amplification security margin produced the expected monotonic reduction in usable secret material. For benign sessions, the mean final secret decreased from approximately 1409 bits at a 64-bit margin to 1338, 1269 and 1173 bits at margins of 128, 192 and 256 bits, respectively. The selected 128-bit baseline therefore introduces additional security compression while retaining sufficient entropy for subsequent HKDF-based session derivation.

E. Reconciliation Robustness

The improved Cascade implementation remains highly reliable across the complete OFAT evaluation. Fig. 7 summarizes the residual reconciliation failures observed for each experimental group after introducing full recursive look-back. No reconciliation failure was recorded for the transmission batch-size study, while only a single failure occurred in the Eve-interception, privacy-security-margin and parameter-estimation groups. The QBER-threshold, basis-bias and distance studies produced only 2, 3 and 4 residual failures, respectively, despite each group containing hundreds of independently executed protocol trials.

These results indicate that the recursive look-back mechanism successfully eliminated the large majority of hidden residual bit disagreements that affected the earlier simplified reconciliation implementation. The small increase in failures observed for basis bias and communication distance is consistent with those experiments producing more challenging post-QKD key conditions, particularly at extreme basis probabilities and long distances where statistical imbalance and physical-channel noise increase. Importantly, reconciliation failures remained rare relative to the total number of executed

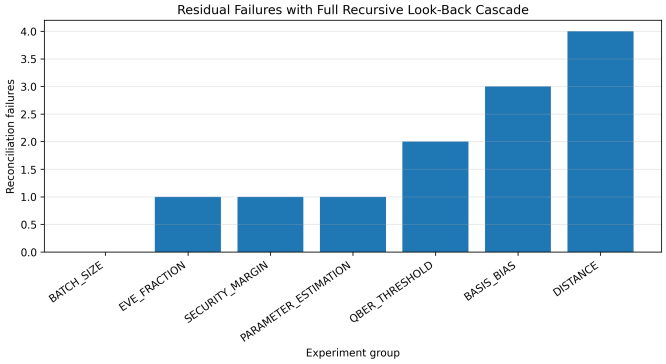


Fig. 7. Residual reconciliation failures across OFAT experiment groups using the full recursive look-back Cascade implementation.

trials, demonstrating that information reconciliation no longer represents the dominant reliability bottleneck of the proposed framework.

F. Overall Discussion

Taken together, the OFAT results reveal that the principal limitation of the proposed framework is not the classical cryptographic pipeline but the physical cost of quantum communication. Around the baseline configuration, variations in batch size, parameter-estimation fraction and QBER threshold preserve near-complete legitimate-session reliability. In contrast, communication distance causes an exponential growth in required optical transmissions, while adversarial interception directly reduces both session success and recoverable secret material. Basis bias further demonstrates that maximizing raw key efficiency is not necessarily equivalent to maximizing security observability.

These results support the use of the selected baseline as a balanced operating configuration for the present prototype: $P(Z) = 0.8$ improves sifted-key efficiency without exhibiting the adversarial weakness observed at extreme bias, a 15% parameter-estimation fraction provides substantial sampling while preserving usable key material, an 8% QBER threshold remains well separated from the approximately 1% benign operating region, and a 128-bit privacy margin preserves sufficient final secret material for session-key derivation. The experiments therefore demonstrate not merely that the integrated pipeline can execute successfully, but that its individual security and physical-layer parameters produce measurable and interpretable effects consistent with their intended protocol roles.

IX. Conclusion

This work shows an end-to-end post quantum secured and quantum secured biometric authentication and communication system which integrates local biometric reconstruction, ML-DSA based device and user authentication, QAuth based session authorisation, biased basis Quantum Key Distribution, recursive Cascade key reconciliation, privacy enhancement and an AES-GCM protected communication within a single dedicated pipeline. This architecture is made to avoid transmission of reusable biometric templates and providing protection against authentication and session establishment against classical and quantum capable adversarial attacks.

The experimental evaluation conducted as a part of this paper demonstrates predictable behaviour of the channel variations. Under the validated baseline configuration, legitimate sessions are highly reliable across communication distances of upto $150km$. The OFAT study also showed that aggressive basis bias improves sifted-key efficiency at the cost of weaker disturbance observability, while increasing Eve interception progressively reduces both session success and usable secret material. These results indicate that the proposed framework is not only functionally integrated, but also exposes measurable security–efficiency trade-offs that must be considered during deployment.

X. Future Work

Many extensions are required in this current study before the proposed prototype can be considered suitable for a practical deployment. First and foremost, the synthetic iris representation should be replaced with data acquired from real biometric sensors and evaluate fuzzy reconstruction under realistic biometric parameters. The trusted terminal should also be made physical attack proof by hardware backed execution, secure memory handling, resistance to side channels and physical extraction attacks.

On the quantum communication side, future work should evaluate the protocol using experimental optical hardware and real detector characteristics rather than a simulated physical channel. Extending the architecture beyond point-to-point communication will additionally require scalable QKD networking, trusted or quantum repeater support and distributed key-management strategies. Finally, the present one-time authentication model may be extended toward continuous biometric authentication, allowing the system to periodically revalidate user presence while preserving an already established secure session. These directions would transform the current research prototype into a more realistic foundation for deployable quantum-secured biometric systems.

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